

1. Administrative & Study Identification

Full Study Title: A Two-Part, Seamless, Multi-Center, Randomized, Placebo-Controlled, Double-Blind Study to Investigate the Safety, Tolerability, Pharmacokinetics, Pharmacodynamics and Efficacy of R07204239 in Combination With Risdiplam (R07034067) in Patients With Spinal Muscular Atrophy **Short Title/Acronym:** MANATEE
Protocol Number: BN42644 (2023-506761-65-00) **NCT Number:** NCT05115110 **Sponsor Name:** Hoffmann-La Roche (Company: Roche)
Trial Status: Active, not recruiting **Investigational Product:** R07204239 (GYM329, anti-myostatin antibody) in combination with Risdiplam (R07034067 / Trade Name: Evrysdi) **Phase of Development:** Phase 2 / Phase 3 **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To investigate the safety, tolerability, pharmacokinetics, pharmacodynamics, and efficacy of R07204239 in combination with risdiplam in participants with Spinal Muscular Atrophy (SMA). Part 1 focuses on safety and dose-finding, while Part 2 focuses on evaluating the combination's clinical efficacy. **Secondary Objectives:** To evaluate how the body processes R07204239 and risdiplam (PK/PD), evaluate motor function improvements, and assess changes in muscle size and strength.

2.2 Background & Rationale Risdiplam works by helping the body produce more survival motor neuron (SMN) protein, which prevents the loss of motor neurons and improves muscle function in patients with SMA. R07204239 is an investigational anti-myostatin antibody designed to target myostatin, a protein that regulates and limits skeletal muscle growth. Inhibiting myostatin may help muscles grow in size and strength. The combination of R07204239 and risdiplam has the potential to deliver a complementary effect, further improving motor function and clinical outcomes for people living with SMA compared to standard SMN-targeted therapy alone.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled (R07204239 or Placebo, alongside background Risdiplam) **Blinding:** Double-Blind **Randomization:** Randomized
Status: Active, not recruiting

3.2 Planned Enrollment & Duration Target \$\$\$: 259 participants
Number of Sites: Multi-center global clinical trial **Estimated Study Start:** 2022 **Estimated Primary Completion:** Jun 27, 2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age at screening: - Part 1 Cohorts A, B, and D: 5-10 years, inclusive. - Part 1 Cohort C: 2-4 years, inclusive. - Part 2 (ambulant participants): 2-25 years, inclusive. 2. Confirmed genetic diagnosis of 5q-autosomal recessive SMA. 3. Symptomatic SMA disease, as per investigator's clinical judgement. 4. Participants who have received previous SMA disease-modifying therapies may be included provided that: - Onasemnogene abeparvovec was received at least 90 days prior to screening, participants are tapered off steroids, and have normal liver function tests, coagulatory parameters, platelets, and troponin-I 90 days post-dose or 1 month after tapering corticosteroids (whichever is later). - Nusinersen last dose was received at least 90 days prior to screening. - Risdiplam is switched to the investigational medicinal product (IMP) provided by the site. 5. Ambulant Participants (Part 1 Cohorts A, B, C and Part 2): Able to walk/run unassisted 10 meters in ≤ 30 seconds (Timed 10-Meter Walk/Run Test [10MWRT]). 6. Non-ambulant Participants (Part 1 Cohort D): Able to sit (MFM32 Item 9 score of 3; MFM32 Item 10 score of at least 2) and able to raise a standardized plastic cup with a 200g weight to the mouth (score of 3 on the entry item of the Revised Upper Limb Module [RULM]).

4.2 Exclusion Criteria 1. Concomitant or previous participation in any investigational drug or device study within 90 days prior to screening or 5 half-lives of the drug (whichever is longer), except completed risdiplam, nusinersen, or onasemnogene abeparvovec studies. 2. Receiving or previous administration of anti-myostatin therapies or any history of cell therapy. 3. Hospitalization for a pulmonary event within the last 2 months or planned hospitalization at screening. 4. Past surgery for scoliosis or hip fixation in the 6 months preceding screening, or planned within the next 9 months (Part 1) or 21 months (Part 2). 5. Unstable gastrointestinal, renal, hepatic, endocrine, or cardiovascular system diseases considered clinically significant, including clinically significant ECG abnormalities at screening, abnormal findings at echocardiography, or cardiovascular disease indicating safety risk. 6. Any major illness within 1 month before screening. 7. Received any multidrug and toxin extrusion (MATE1/2K) substrates within 2 weeks before screening. 8. Hereditary fructose intolerance. 9. Used specified medications within 90 days prior to screening (e.g., riluzole, valproic acid, hydroxyurea, sodium phenylbutyrate, butyrate derivatives, creatine, carnitine, growth hormone, anabolic steroids, probenecid, acetyl cholinesterase inhibitors, agents with known/presumed HDAC inhibitory effect). 10. Ascertained or presumptive hypersensitivity to R07204239 or risdiplam. 11. Clinically

relevant history of anaphylactic reaction requiring inotropic support. 12. Any abnormal skin conditions, pigmentation, or lesions in the abdomen preventing visualization of potential subcutaneous injection site reactions to R07204239. 13. Immobilization, surgical procedures, fracture, or trauma to limbs within 90 days prior to screening. 14. **Part 1 Cohorts A and B only:** Contraindications for MRI scan (e.g., claustrophobia, pacemakers, metal implants, spinal rods). 15. **Part 1 Cohort D only:** Inability to adopt position for DXA scan; contractures interfering with DXA/functional assessments; able to walk unassisted (10m in ≤ 30 s); severe scoliosis (curvature $> 40^\circ$); requirement for invasive ventilation, tracheostomy, or use of daytime noninvasive ventilation.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Risdiplam is administered as an approved daily dose. R07204239 (or placebo) is administered every 4 weeks. **Route of Administration:** Risdiplam (Oral) + R07204239/Placebo (Intravenous/Subcutaneous Injection). **Duration of Treatment:** Up to approximately 4 years of total participation, including the extension phase (Part 1 core is 6 months; Part 2 core is 18 months).

5.2 Concomitant & Prohibited Therapy Permitted: Standard of care for SMA; participants receive risdiplam background therapy for at least 8 weeks prior to combination. **Prohibited:** Other SMA disease-modifying therapies within strict washout periods (onasemnogene abeparvovec < 90 days, nusinersen < 90 days), previous anti-myostatin therapies, cell therapy, and specific agents affecting muscle strength or HDAC inhibitors within 90 days.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Genetic Confirmation, Medical History, ECG/Echocardiography, Baseline Motor Function, MRI/DXA (cohort dependent)
Baseline	Day 0	Randomization, First Doses of R07204239/Placebo (with daily Risdiplam)
Interim	Every 4 Weeks	Safety Labs, R07204239 Administration, PK/PD Assessment, Motor Function Scales
End of Study	Follow-Up	Follow-up appointments 3 months and 6 months after the last dose of trial treatment

7. Outcome Measures (Endpoints)

7.1 Primary Outcomes 1. **Part 1** - Percentage of participants with adverse events (AEs). 2. **Part 1** - Incidence of relevant

echocardiographic parameter z scores > 2. 3. **Part 1** - Serum concentration of R07204239.

7.2 Secondary Outcomes 1. **Part 2** - Change from baseline in Motor Function Measure (MFM) Domain 1 + Domain 2 (D1 + D2) score. 2. **Part 2** - Change from baseline in MFM-32 total score. 3. **Part 2** - Change from baseline in time taken to rise from the floor as measured by RHS Item 25.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine collection via onsite visits every 4 weeks, with specific attention to liver function tests, coagulation parameters, platelets, echocardiography parameters, and troponin-I. **Serious Adverse Events (SAEs):** Standard reporting timeline, typically within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** External independent Data Monitoring Committee responsible for ongoing safety reviews, especially during the Part 1 dose-escalation phase.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy outcomes in Part 2. Safety Set for all participants receiving at least one dose of the investigational product. **Sample Size Justification:** Target $N = 259$ is appropriately powered to detect significant differences in safety and efficacy, divided into Part 1 dose-finding and Part 2 efficacy confirmation. **Handling of Missing Data:** Standard Mixed-Effect Model Repeated Measure (MMRM) and/or Multiple Imputation for longitudinal motor function efficacy endpoints.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite monitoring visits combined with centralized data reviews to ensure protocol adherence. **Audit Trail:** Secure eCRF systems with automated audit trails and data clarification form workflows.

11. Study Identifiers & Governance

Primary ID: BN42644 **Registry Number:** NCT05115110 **Sponsor/Lead Organization:** Hoffmann-La Roche **Study Title:** MANATEE **Official Regulatory Title:** A Two-Part, Seamless, Multi-Center, Randomized, Placebo-Controlled, Double-Blind Study to Investigate the Safety,

Tolerability, Pharmacokinetics, Pharmacodynamics and Efficacy of R07204239 in Combination With Risdiplam (R07034067) in Patients With Spinal Muscular Atrophy **MeSH Code:** D013110 **SNOMED ID:** 196562006 **ATC Code:** M09AX10

12. Clinical Framework (SMA Specific)

Primary Condition: Spinal Muscular Atrophy (SMA) **Diagnosis Threshold:** Confirmed genetic diagnosis of 5q-autosomal recessive SMA **Medical Methodology:** Combined SMN-enhancing therapy (Risdiplam) + Anti-myostatin skeletal muscle growth stimulation (R07204239) **Optimization Target:** Clinically meaningful improvements in motor milestones and skeletal muscle strength

13. Study Procedures & Methodology

Management Pathway: Standard clinical administration protocol for combination biological therapy in SMA. **Education Protocol (HCP):** Site initiation training for the administration and tracking of R07204239 alongside daily oral risdiplam. **Education Protocol (Patient):** Comprehensive informed consent detailing the multi-year trial duration and double-blind placebo component. **Quality Audit Mechanism:** Real-time data entry tracking and source data verification.

14. Observation & Follow-Up Schedule

Total Duration: Approximately 4 years (including long-term open-label extension phase) **Onsite Visit Cadence:** Every 4 weeks for investigational product administration and assessments. **Remote Monitoring Frequency:** Supplementary safety calls as indicated by the protocol. **Checklist Review Interval:** Clinical milestone and safety reviews performed concurrently with the Q4W dosing visits.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				